

Programming Fundamental

Lab 11

objective:

- To understand the fundamental difference between stack and heap memory in program execution.
- To demonstrate how variables are stored in stack vs. heap memory.
- To implement an array using heap memory and observe how it regrows as more elements are inserted.

Task 1: Use heap allocation to read integers from a file. Create a function to count how many of the numbers are prime and display the result.

Sample File (data.txt):

10 2 3 4 5 6 7 8 9 10 11

Sample Output:

Total Prime Numbers: 5

Task 2: Reverse File Data

Objective: Read integers from a file into a heap-allocated array. Create a function to reverse the array contents in-place and display both original and reversed arrays.

Sample File (data.txt):

6 10 20 30 40 50 60

Sample Output:

```
yaml
CopyEdit
Original Data: 10 20 30 40 50 60
Reversed Data: 60 50 40 30 20 10
```

Task 3: Find Second Maximum

Objective: Use heap memory to read an array from a file. Implement a function that finds the second largest number in the array.

Sample File (data.txt):

7 15 42 23 8 37 42 10

Sample Output:

Second Maximum: 37

Task 4: Remove Duplicates

Objective: Read integers from a file into a heap-allocated array. Create a regrowable array to store only unique elements (removing duplicates). Output the modified array.

Sample File (data.txt):

10 4 5 4 6 5 7 8 8 9 10

Sample Output:

Modified Data (Unique): 4 5 6 7 8 9 10

Task 5: Character Frequency Count

Objective: Read characters from a text file into a heap array. Count and display the frequency of each vowel (a, e, i, o, u) using a function.

Sample File (text.txt):

education is important

Sample Output:

```
makefile
CopyEdit
a: 2
e: 1
i: 2
o: 1
u: 1
```

Task 6: Filter Palindromic Words

Objective: Read words from a file using dynamic heap allocation. Identify and store palindromes into a regrowable array and print them.

Sample File (text.txt):

LEVEL HELLO CIVIC CODE RADAR WORLD

Sample Output:

Palindromic Words: LEVEL CIVIC RADAR